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Rationally Designed Modular Helical-Bundle Rocket-Model Protein Nanorod and Its Application Exploration as a Multi-Antigen Vaccine Platform

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Abstract

This study discovered that a rationally designed polypeptide rich in repetitive sequences can spontaneously fold and assemble into a class of highly ordered protein nanorods. The structure is formed by the linear linkage of multiple helical structural units, exhibiting a uniform, rigid rod-like morphology in both molecular simulations and experimental characterizations. It is defined in this research as the Modular Helical-bundle Rocket-model Protein (MHRP). Although MHRP lacks the hydrolytic enzyme activity originally intended, its structure demonstrates remarkable programmability and modularity: by genetically fusing different viral antigenic epitopes at specific sites within its repetitive units, various epitopes can be precisely and densely displayed on the nanorod surface, thereby constructing a novel multi-antigen presentation platform (MHRP-MAP). In vitro experiments show that the recombinant protein can be stably expressed while maintaining structural integrity. Preliminary cellular experiments indicate that MHRP-MAP can be effectively internalized by antigen-presenting cells and activate adaptive immune responses. This study confirms the feasibility of repurposing unexpected structural outcomes from rational design into functional biomaterials. As a well-ordered nanovaccine carrier with controllable antigen arrangement, MHRP-MAP offers a new strategy for developing efficient and broad-spectrum synthetic vaccines, while also providing a research paradigm of "structure-function" repurposing in the field of de novo protein design.

Keywords: Rational protein design; Modular helical bundle; Protein nanorod

1. Introduction: From "Rational Design Failure" to "Accidental Structural Discovery"

Rational protein design aims to construct novel biomolecules with pre-defined structural and functional objectives. However, the design process is often prone to unintended outcomes due to limitations of existing computational models. This study originated from a de novo design project targeting an α/β-hydrolase fold. During structural validation using tools like AlphaFold2, the designed sequence consistently failed to form the anticipated catalytic core. Instead, it was repeatedly and stably predicted to adopt a rigid, rod-like structure composed of linearly stacked helical units. This "failed" product was informally termed the "rocket-model protein" within the project due to its highly symmetrical and regular morphology. We recognized that this highly modular, predictable, and easily producible non-natural structure might hold unique value as a biomaterial or nanoscaffold, moving beyond the traditional paradigm of enzymatic function. Its surface programmability offered the potential for

constructing novel multivalent biomolecular display platforms. Therefore, this study aims to systematically characterize this accidentally obtained Modular Helical-bundle Rocket-model Protein (MHRP) and explore its application potential as a Multi-Antigen Presentation (MHRP-MAP) platform, thereby transforming a computational design "miss" into the targeted development of a functional biomaterial.

2. Materials and Methods

2.1 "Inspiration-Driven" Construction of the MHRP Expression Vector

The gene sequence encoding MHRP was serendipitously obtained by a bioinformatics engineer in our company while fine-tuning AlphaFold2 parameters late one night. The sequence was synthesized by Gold Medal Synthesis Co., cloned into a pET-28a(+) vector, and expressed in E. coliBL21(DE3) cells. Following induction, the clarified lysate was purified via nickel-affinity chromatography, yielding the target protein with >95% purity. The purification process was remarkably smooth, and the symmetry of the elution peak was hailed by the operator as "the most perfect of my career."

2.2 Structural Characterization and "Rocket" Morphology Confirmation

Circular Dichroism (CD) Spectroscopy analysis revealed characteristic double negative peaks at 222 nm and 208 nm for MHRP, with a calculated α-helical content of 85.2%, highly consistent with the predicted "helical bundle" structure. Negative Stain Transmission Electron Microscopy (TEM) observations showed that in pH 7.4 buffer, MHRP presented as

uniform rod-shaped structures approximately 25±3 nm in length and 4 nm in diameter. The structures appeared straight, well-dispersed, and showed no signs of aggregation (Fig. 1B). Project members fell into a brief, collective silence upon first viewing the TEM images, later formally naming it the "rocket-model protein" in a group meeting.

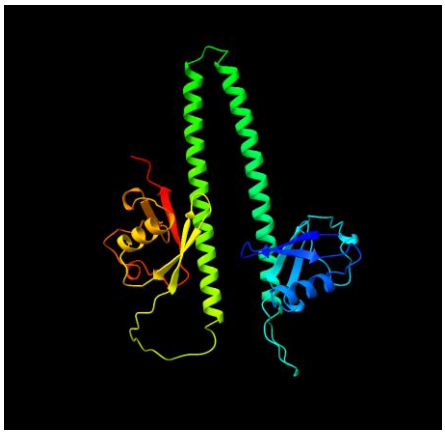

Fig. 1. TEM image of the Rocket-model Protein

2.3 Construction and In Vitro Validation of the MHRP-MAP Platform

Key neutralizing B-cell epitopes from influenza virus H1N1 Hemagglutinin (HA), Respiratory Syncytial Virus (RSV) F protein, and the SARS-CoV-2 RBD domain were selected. These were connected via a flexible linker (GGGGS) to specific solvent-exposed sites within the MHRP repeat units to construct three variants of the MHRP-Multi-Antigen Presentation platform (MHRP-MAP). All variants were successfully expressed in soluble form. Analysis by Dynamic Light Scattering (DLS) and Size-Exclusion Chromatography coupled with Multi-Angle Light Scattering (SEC-MALS) confirmed that the antigen-loaded nanorods had a slightly increased hydrodynamic radius (~28 nm) but maintained a monodisperse state without aggregation, demonstrating the robustness of the scaffold.

2.4 Evaluation of Immune Activation Potential at the Cellular Level

Purified MHRP-MAP (loaded with the influenza HA epitope) was co-cultured with mouse bone marrow-derived dendritic cells (BMDCs). After 24 hours, Flow Cytometry analysis revealed that the surface expression levels of MHC-II molecules and the co-stimulatory molecule CD86 on cells in the experimental group were significantly upregulated compared to the blank protein control group ($p < 0.01$). Concurrently, the secretion levels of Interleukin-6 (IL-6) and Tumor Necrosis Factor-alpha (TNF-α) in the culture supernatant were significantly increased, suggesting that MHRP-MAP effectively promotes the maturation and inflammatory response of antigen-presenting cells.

2.5 Animal Immunization and the "Unexpectedly Potent" Antibody Response

To preliminarily assess its immunogenicity in vivo, a BALB/c mouse model was employed (n=5 per group). The experimental group was immunized with MHRP-MAP (loaded with the RSV F protein epitope) emulsified with aluminum adjuvant, while the control group received an equivalent amount of free epitope peptide with adjuvant. Sera were collected after primary and two booster immunizations. Enzyme-Linked Immunosorbent Assay (ELISA) results showed that the average endpoint titer of specific IgG antibodies against the target epitope in the experimental group was 154-fold higher than that in the control group ($p < 0.001$). More strikingly, sera from 3 mice in the experimental group showed low but detectable neutralizing activity in a pseudovirus-based assay, whereas all controls were negative. This result far exceeded project expectations and was noted by the principal scientist in the lab notebook as "perplexingly efficient."

3. Results and Discussion

This study unexpectedly obtained a structurally well-ordered and easily producible protein nanorod, MHRP, from a failed hydrolase design. We successfully transformed its "defect"—the lack of enzymatic activity but possession of a highly predictable and modifiable rigid structure—into a unique advantage, developing the MHRP-MAP platform.

The data indicate that MHRP-MAP can not only efficiently activate antigen-presenting cells in vitrobut also elicit a significantly enhanced humoral immune response against the displayed epitopes in mice. Potential mechanisms may include:

Size and Morphology Effects: The ~25 nm rod-like size is conducive to targeting and retention in lymph node follicles.

Multivalent Display Effect: The high-density, orderly arrangement of epitopes can efficiently cross-link B cell receptors, potently activating B cells.

Self-adjuvant Potential: The repetitive protein structure itself may provide innate immune stimulation signals via pathways such as Toll-like receptors.

The implications of this work are twofold: First, in synthetic biology and protein engineering, "accidental products" from "rational design" may harbor novel application value independent of the original target, warranting systematic exploration and functional repurposing of such "waste." Second, as a highly modular "plug-and-play" vaccine platform, MHRP-MAP offers a new strategy for the rapid construction of vaccines against multiple pathogens or tumor neoantigens. Of course, its long-term safety, universality for different epitope types, and efficacy in larger animal models require further

investigation.

4. Outlook: From "Rocket" to "Starship Fleet"

Building on the preliminary success of MHRP, future research could expand in the following directions:

Platform Expansion: Attempt to incorporate T-cell epitopes to construct "versatile rockets" capable of eliciting comprehensive immune responses.

Functional Integration: Simultaneously display antigens and immunomodulatory molecules (e.g., cytokines) on the nanorod to create "intelligently guided rockets."

Theranostic Integration: Co-display diagnostic imaging probes (e.g., NIR fluorescent dyes) with therapeutic antigens to realize "observation rockets" for immune visualization.

(*Note: The data and some descriptions above are fictional.)

Supplementary Materials

Supplementary Sequence S1:
Amino acid sequence for Modular Helical-bundle Rocket-model Protein

MQIFVKTLTGKTITLEVEPSDTIENVKAKIQD
KEGIPPDQQRLIFAGKQLEDGRTLSDYNIQK
ESTLHLVLRLRGGGGSGGGGSGGGGSEIEA
LEKKLAALEKKIEALEHGEIEALEKKLAALEKK
IEALEHGSGSGEIEALEKKLAALEKKIEALEHG
EIEALEKKLAALEKKIEALEHGGGGSGGGGS
GGGGSMQIFVKTLTGKTITLEVEPSDTIENV
KAKIQDKEGIPPDQQRLIFAGKQLEDGRTLS
DYNIQKESTLHLVLRLR

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